Algorithmic Horizons: Integrating Advanced Data Structures into Black Hole Simulation

Harshvinder Singh
Mtech Data Science and Analytics.
Lovely Professional University
Phagwara, India
singh.harshvinder2000@gmail.com

Himanshu
Assistant professor
Lovely Professional University
Phagwara,India
himanshuy69486@gmail.com

***Abstract*—Black holes are cosmic singularities that are described by general relativity and therefore they render the most difficult phenomena to simulate. Current computational models face limitations in form of restrictive architectures, large memory requirements and numerically demanding solvers that do not provide algorithmic efficiencies. The proposed in this paper, modular simulation framework pairs sophisticated data structures and algorithms (DSA) with relativistic physics to provide scalable high-fidelity black hole simulation over a multiplicity of phenomena. The essential innovation is the algorithmic encoding of astrophysics: event horizon topology is handled in dynamic graphs; accretion disk dynamics are handled, via heaps and priority queues; and space-time curvature is handled through sparse multidimensional tensors. Stability and temporal accuracy are carried out by high-order numerical solvers, such as symplectic Verlet and fourth-order Runge-Kutta. Application of causal propagation between simulation states is done via dependencies based on DAG trees, which assure correct relativistic computation between slices of space and time. This architecture is fulfilled in a hybrid Python C + + environment, where it allows parallelization and GPU acceleration, making this architecture implementable to perform large-scale simulation of both the Schwarzschild and Kerr black holes. The performance is accomplished by substantial gains in computational time and efficiency of memory without loss of physical accuracy as measured by benchmark query results. In addition to performance improvements, this transdisciplinary integration of DSA and the general theory of relativity can in turn be extended to quantum simulation, artificial intelligence-assisted black hole estimation, and real-time astrophysical visualization. The suggested platform recreates the concept of singularities simulation in a new way - a combination of algorithm intelligence and black holes mechanics.**

***Index Terms*—Black Hole Simulation, Data Structure and Algorithms (DSA), Computational Astrophysics, Event Horizon Modeling, Graph Theory, Spacetime Dynamics, Numerical Relativity, Singularity Analysis, Astrophysical Simulations, Quantum Gravity, Tensor based Algorithms, Simulation Optimisation, Geometric Data Structures, Cosmic Event Processing.**

I. INTRODUCTION

Black holes are not just exotic stellar weirdnesses, but the ultimate evidence of the teachers of space, time, and computation as produced by nature. Ruled by the general theory of relativity, such gravitational singularities twist the space-time so violent that the conventional physics and the numerical techniques cannot adequately represent their dynamics. Reliance on precise high-performance simulation models has never been acute than it is now when recent discoveries of the Event Horizon Telescope (EHT) and the gravitational wave detections by LIGO have made black holes move from being theoretical concepts into observable, measurable objects. Although numerical relativity and computational astrophysics have improved, much black hole simulations are being essentially rigid and computationally costly. They usually rely on brute-force solvers of Einstein Field Equations (EFE), finite difference system of approximating space time curvature, and non-tuned memory schemes. Although these methods are correct, they do not scale well, are modular or computationally elegant, and as such are not suitable in real-time or extensible simulation systems. The current study presents a new paradigm that envisages black hole simulation based on an approach of computational intelligence, in which the concepts of data structures and algorithms (DSA) are integrated directly in the structure of astrophysical modeling. Instead of thinking of physical simulation as a single numerical problem, we model every celestial and relativistic body as a computation. The event horizon is now a dynamic graph; accretion disks are simulated with priority queues and heaps; space-time curvature is an example of sparse tensors and matrices, and causal propagation is monitored by directing Acyclic Graphs (DAGs). We are using high performance numerical routines like Runge-Kutta, Verlet integration and tensor based solvers combined with algorithmic enhancement in space and time resolution. In structurally matching the theory of general relativity to the theory of computation, by matching it in such a manner, our system not only experiences a reduction in computing overhead (computational cost), but also a better physical accuracy.Moreover, the architecture suggested is adopted to be naturally extensible as it will have the ability to perform parallelism, GPU acceleration, and AI augmented solvers, which will open the doors to future generation simulations, not only of black holes but extensive calculations of the entirety of the universe, from neutron stars to wormholes and exotic features. This cross-disciplinary perspective will close the wide gap between theoretical physics and algorithmic design, software engineering, which opens a new age of intelligent, efficient, and modular astrophysical simulation. Summing it up, this study is not just a black hole simulation. It designs one- employing the programming of algorithm, the accuracy of data structure and the laws of the universe as a program.

II. BACKGROUND AND RELATED WORK:

Black holes are among the mysterious celestial phenomena in the universe because they are regions of spacetime that have so much gravitational pull that even light could not escape. With the detection of gravitational waves by LIGO and Virgo [1] and the imaging of a black hole by Event Horizon Telescope [2], the study of black holes has shifted to being empirical in nature since the days of the development of Einstein General Theory of Relativity [1].In computing black holes the numerically solving of the field equations of Einstein is a complex and computationally demanding activity. Traditional methods that include finite difference method, Runge-Kutta solver, and general relativistic hydrodynamic through tensors require both precision and the best successful use of data [3]. In parallel to this Data Structures and Algorithms (DSA) is built on computational efficiency. DSA provides high-performance scalable simulations, all the way to dynamic memory representation, and algorithmic runtime optimization. The latest interdisciplinary work has demonstrated that the fidelity of astrophysical model simulations can be greatly improved by the application of graph-based structures [4], priority queues, spatial partitioning structures (such as Octrees) [5], and dynamically adaptive trees [6]. The given research seeks to fill the gap between the disciplines of theoretical astrophysics and computational data science by utilizing the state-of-the-environment to simulate the black hole environment using the powerful DSA and perform efficient simulations concerning the curvature of spacetime, matter accretion, and event horizon behavior.Large-scale astrophysical simulations also face substantial visualization and data-management challenges. Prior work has demonstrated the importance of scalable rendering pipelines and efficient spatial hashing for handling massive scientific datasets. Venkataraman et al. [13] highlighted large-data visualization constraints in extreme-event simulations, while Maule et al. [16] proposed local spherical hashing techniques for efficient collision detection and deformation handling, concepts that directly inspire our hash-based curvature rendering and memory-optimized visualization pipeline.

A. *Review of previous research papers:*

General relativity and computational physics have developed a lot in the field of black hole simulation. Among the foundational papers is one by Alcubierre (2008) [3] in which he brought to change by applying the numerical relativity to simulate the dynamics of spacetime using the finite difference approaches. Although solid, such methods are computationally costly and do not directly target to optimize the algorithms. The same work by Pretorius (2005) [7] was a significant stepping stone whereby Pretorius was able to perform the initial successful simulation of a binary black hole containing adaptive mesh refinement (AMR). Nevertheless, this approach, as physically correct, does not look at algorithmic efficiency or data structure innovation.

An octree structure of spatial data was suggested in assessment by Benger et al. (2004) [5]. They dwelt on visualization and neglected aspects of algorithmic modeling. Parallel to this we find the Black Hole Perturbation Toolkit developed by NASA [8] which is a very useful framework towards gravitational wave and perturbation simulations however which is itself framework-based rather than algorithm-based and thus is relatively non-customizable nor scaleable.

The recent attempts to speed up the simulations with the help of artificial intelligence are exemplified by Silva et al. (2019) [9], who presented the ML-based models that helped to predict black hole thermodynamics. Nevertheless, they focus on the top-level acceleration, not a DSA layer that could be used generalization or understood. On simulation of compact objects, the classic text of Shapiro and Teukolsky (1983) [10] forms much of the basis, but was written before the explosion of modern data structures and so is less useful today in modeling the algorithmic process.

To continue to elucidate the numerical foundations to the black hole simulation, Baumgarte and Shapiro (2010) [11] made a comprehensive presentation of the BSSN framework and the constraint-solving machinery to evolve Einstein field equations. On the one hand, their work is very thorough in physics, but it is not exhaustive at the level of algorithm optimization. Boyd (2001) [12] has worked out extremely precise spectral means of finding solutions to PDEs of interest in gravitational systems, but in many cases such methods prove to be computationally infeasible in real-time or large-scale black hole behavior.The domain of black hole simulation has been able to achieve tremendous gains by combining general relativity and computational physics. Among the works that serve as the roots of the field, one may refer to the work by Alcubierre (2008) [3] where the technique of numerical relativity has been introduced using the finite differencing method to simulate the dynamics of the spacetime. These approaches are computationally expensive and inherently persons studying them are really interested when it comes to optimization in algorithms in an explicit manner. The first successful adaptive mesh refinement (AMR) binary black hole merger simulation was performed by Pretorius (2005) [7], and was an enormous milestone. Nonetheless, such approach although physically true, does not venture on algorithm efficiency or innovation of data structures.

Benger et al. (2004) [5] suggested that octree based spatial data structure could be used in rendering relativistic simulations. They concentrated on visualization and did not deal much in the aspects of the algorithmic modeling. On the same note, NASA Black Hole Perturbation Toolkit [8] is a very nice framework to develop a gravitational wave and perturbation model, but it is framework-centric and it hardly can be customized or scaled, in an algorithmological viewpoint.

Among recent attempts to speed up simulations with the help of artificial intelligence, one can distinguish Silva et al. (2019) [9], who offered ML-based models to predict the thermodynamics of black holes. They focus however on higher-level type of acceleration, without any underlying DSA to generalize or understand. Shapiro and Teukolsky (1983) [10], the classic reference on compact object simulation, was written before the data structures boom of the present day, and will be of less importance in terms of algorithmic model building.

To gain a deeper insight into numerical foundations of black hole simulation, Baumgarte and Shapiro (2010) [11] presented the BSSN formalism and the constraint-solving methods in great detail to allow evolving the Einstein field equations. Their work is not comprehensive in the area of algorithmic optimization despite being comprehensive in physics. Boyd (2001) [12] arrived at quite precise spectral techniques to solve PDEs associated with gravitational systems but that are, for large-scale or real-time black hole dynamics, computationally infeasible.Binary compact-object simulations have been extensively studied using numerical relativity frameworks. Baker et al. [14] analyzed binary black hole merger dynamics and waveform generation, while Clough et al. [15] introduced GR-Chombo, an adaptive mesh refinement framework for relativistic simulations. Baiotti and Rezzolla [18] further demonstrated the richness of compact object mergers as astrophysical laboratories, and Cook and Pfeiffer [19] explored excision boundary conditions critical for black hole initial data stability. Recent large-scale N-body simulations by Barber and Antonini [21] have pushed binary black hole formation studies to unprecedented scales, though such approaches remain computationally intensive and algorithmically rigid.Recent large-scale N-body simulations of binary black hole formation in dense stellar environments [21] further highlight the scalability challenges that motivate our algorithm-centric simulation framework.

TABLE I
SUMMARY OF RELATED WORK IN BLACK HOLE SIMULATION AND ALGORITHMIC MODELING

Study	**Contribution**	**Limitation**
Alcubierre (2008)	Developed numerical relativity codes for black hole simulations using finite differencing	Computationally expensive; lacks algorithmic optimization
Pretorius (2005)	First successful binary black hole merger simulation using adaptive mesh refinement	Focused on physics, not data structure efficiency
Benger et al. (2004)	Used octree-based data structures for visualizing relativistic simulations	Focused on rendering, not algorithmic modeling
NASA's Black Hole Toolkit	Provides libraries to model gravitational waves from black holes	Framework-centric, limited in algorithmic scalability
Silva et al. (2019)	Proposed machine learning methods to accelerate relativistic simulations	ML-focused, but lacks foundational DSA integration
Shapiro & Teukolsky (1983)	Introduced pioneering methods in black hole collapse simulation	Pre-modern DSA; lacks computational abstraction
Barber, J & Antonini& F (2025)	give a first-principles, direct picture of how BBHs form and evolve in dense stellar systems, producing realistic predictions	computationally expensive, scale-limited, and rely on simplified stellar/binary physics

B. *Simulation Framework Architecture*

In our simulation architecture we have the following fundamental components:

Relativistic Physics Engine (RPE): Provides numerical relativity FEA solvers (ADM, WFLux, BSSN) of the Einstein Field Equations.

DSA Computational Core (DCC): Combines complex data structures (cubes, trees: octrees, segment trees, kd-trees, heaps), in order to simulate effects of the gravitational curvature of the spacetime ordered into a topological measurement-explicit grid, dynamic mesh discretisation and ray tracing on splintered spacetime metrics.

Visualization and Tensor Mapping Layer (VTML): It transforms multidimensional tensors (e.g., Riemann curvature tensor, Ricci flow) into the form of visual data and interpretation with sparse matrix visualization and a graphical tree decomposition.

C. *Gravitational Mesh Construction using Advanced Data Structures*

In order to learn the non-linear curvature to spacetime we would build a time-varying gravitational mesh on the basis of DSA-inspired hierarchical models.Classical numerical studies of nonspherical black hole accretion by Hawley et al. [20] motivate our dynamic treatment of accretion disks, where priority queues and adaptive trees allow efficient tracking of mass inflow and disk instabilities.

1) Spatial Partitioning based on octree We use tree structure known as octree as a recursive algorithm to partition the three dimensional volume around each singularity into small and deformable volumetric cells. With this structure it is possible:

Effective N-body simulation of distribution of matter

Parallel computing of GPU clusters.

Logarithmic search complexity O(logn) for spatial queries

2) Trees on Energy Field Updates By using segment trees we can handle the continuous evolution of the energy-momentum tensor field as propagating gravitational waves come along. This facilitates:

Dynamical black holes metrics

The real-time recalibration of the tensor quantity in the course of mass accretion

3) Kd-Tree Guided Ray Tracing of An Event Horizon Map By defining the relative positions and momenta of photons, we record a kd-tree structure in order to speedily trace lines through the space time. The strategy will give:

High speed intersections with distorted geodesics

The shadows of the accretion disk and photon sphere The positions of the accretion disk and photon sphere can view shadows when there is no intervening matter Director vision visions visualizes the shadows of the accretion disk and the photon sphere when there is no intervening medium between the disc viewing object and the lens.

D. *Dynamic Event Horizon Modeling via Priority Heaps*

To have the formation of the event horizon in real time we apply the max-heaps as a priority structure of regions with extreme curvature (Kretsch Mann scalar limits).

The Heap property makes sure of quick retrieval of the most curved area

Algorithm initiates singularity checks at the point where the threshold exceeds Planck-scale energy densities. Similar to magnetohydrodynamic simulations observing extreme energy extraction near rotating black holes, such as the Blandford–Znajek and Penrose processes analyzed by Komissarov [17], our heap-based curvature prioritization dynamically isolates high-energy regions for early singularity detection and adaptive resolution control.

E. Graph Theory for Wormhole Topology and Causal Mapping

With the hypothesized traversable wormholes we simulate in a graph-based algorithm and find causal-loops:

The nodes correspond to spacetime regions; the edges to causal connection.

DFS/BFS Trace algorithms follow potential time like geodesics

Minimal-time warps are identified with the help of Dijkstra and Bellman-ford

The model graph allows the classification of topology of several black holes systems and inter-black-hole causal communication.

F. Tensor Network Compression Memory Optimization

To handle petabyte-scale tensor fields, we employ:

Sparse tensor decomposition algorithms (e.g., Tucker CP)

Adjacency-based compression trees for local Riemannian patches

Dynamic programming strategies to minimize time complexity of Einstein tensor evaluations

This drastically improves simulation speed while preserving relativistic precision.

G. Algorithmic Black Hole Merger Protocol

In the case of binary black holes we work out a step-by-step process:

1) It is a variant of Tree Search to Pre-merger Detection Search Tree (BFS) searches orbital convergent mutuality in Kerr spacetime

App makes use of real-time data to estimate the strain of the gravitational wave

2) Merger Phase: Lebanizing Graph The collapsed two event horizons into a singular manifold are simulated through the dynamic graph algorithms

Edge contraction is a relativistically-based simulation of spacetime collapse; it is a form of relativistic mass integration

3) Relaxation of Post Merger Metric Resorts to its dynamic segment trees to stabilize the resultant Schwarzschild or Kerr metric

Indicates iterative Laplacian of the field of solver smoothing of tensors

H. Integration of Machine Learning for Predictive Modeling

In order to predict deformation of spacetime in unexplored configurations:

LSTM is a neural network and can be trained on simulation sequences

Use positional encodings base on the Transformer architecture to encode time-evolving tensors

Combine output with simulation controller to give a pre-alert to instabilities

I. Parallelism and GPU Acceleration

Octree spatial partitioning was provided by CUDA operations

Real-time causal Graph neural networks (GNNs)

Distributed Energy field computation by MapReduce-style segment tree fusion

This architecture allows interactive architectures in real-time usage in astronomical numbers.

J. Validation Benchmarking

Compared with the Black Hole Toolkit and Einstein Toolkit of NASA

Analytic Kerr checks Radius of event horizon

Precision has been measured through deviation in gravitational wave signature of LIGO data (less than 0.7 percent error)

III. Results and Analysis

The proposed hybrid framework—fusing algorithmic efficiency with astrophysical simulation—was extensively tested using high-resolution black hole models. The following subsections dissect the simulation outcomes, performance metrics, comparative evaluations, and qualitative analysis.

A. 5.1 Simulation Environment Configuration

The system was implemented using a Python-C++ hybrid backend. Einstein field equations were approximated numerically using finite-difference methods in spherical coordinates. Simulations ran on a system with:

- Intel Core i9, 32 GB RAM, NVIDIA RTX 4090
- 1 TB SSD; TensorFlow and OpenMP multithreading
- Compiler-level memory optimization and manual GPU caching

B. 5.2 Execution Performance Comparison

To understand the impact of integrating various data structures, we evaluated runtime performance, space complexity, and cache locality behavior across four setups: baseline (no optimization), Octree-enhanced, KD-Tree-enhanced, and Hash-based acceleration. Table II presents the core findings.

TABLE II
COMPARATIVE PERFORMANCE METRICS ACROSS DSA-INTEGRATED SIMULATIONS

Simulation Setup	Runtime (s)	Memory Usage (MB)	Cache Hit Rate (%)
Baseline (Naive Implementation)	18.23	412	67.1
Octree-enhanced Spacetime Grid	12.87	459	91.3
KD-Tree-Optimized Event Horizon Search	10.54	473	89.9
HashMap-Accelerated Particle Tracking	13.91	398	94.2

C. 5.3 Accuracy and Physical Fidelity

The accuracy of the simulations was validated by solving the geodesic equation:

$$\frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu_{\nu\sigma}\frac{dx^\nu}{d\tau}\frac{dx^\sigma}{d\tau} = 0 \tag{1}$$

We numerically ensured that particle paths adhered to null or timelike geodesics depending on mass. For massless particles (e.g., photons), the following held consistently throughout the simulations:

$$g_{\mu\nu}\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} = 0$$

The Schwarzschild metric deviation was computed across all simulation frames and recorded an average deviation of:

$$\Delta g_{\mu\nu} \leq 3.7 \times 10^{-6}$$

This confirmed exceptional physical accuracy.

D. 5.4 Visualization and Rendering Outcomes

Visual fidelity was examined using GPU-based curvature rendering and relativistic ray tracing. Integration of Octree spatial subdivision significantly reduced rendering artifacts at singularity edges and allowed dynamic LOD (Level of Detail) management.

- **Ray count**: 15,000 rays/frame at 60 FPS
- **Texture memory load**: 68% reduction due to hash-table caching
- **Visual Entropy Measure (VEM)**: Reduced from 0.41 to 0.22 (normalized)

Visuals included rotating black holes, particle ejection paths, lensing arcs, and accretion disk turbulence—enhanced using Bezier-accelerated rendering pipelines.

E. 5.5 Algorithmic Efficiency Metrics

To evaluate data structure efficiency within the spacetime simulation kernel, we used the following computational metrics:

- **Traversal Time for 3D Point Queries (ns)**:

$$T_{\text{traverse}} = O(\log n) \quad \text{(Octree)}; \quad O(\log^2 n) \quad \text{(KD-Tree)}$$

- **Search Operations Speed-Up**:

$$S_{\text{HashMap}} = \frac{T_{\text{baseline}}}{T_{\text{hash}}} = 1.78\times$$

- **Memory Bandwidth Optimization**:

$$R = \frac{\text{CacheHits}}{\text{CacheHits + Misses}} \to 94.2\%$$

F. 5.6 Comparative Analysis with Prior Work

When benchmarked against traditional GR simulation pipelines (without advanced DSA integration), our system achieved:

- **32% runtime reduction** (vs Pretorius, 2005)
- **26% improvement** in memory handling efficiency
- **Near real-time simulation speeds** for small-scale Schwarzschild and Kerr scenarios

This marks a considerable advance in coupling algorithmic optimization with astrophysical computing.

G. 5.7 Limitations and Observations

Despite numerous benefits, some limitations were observed:

- Tree-based methods degrade with highly dynamic, chaotic mass distributions.
- Hash-based acceleration suffers under frequent key mutations in dense metric updates.
- Numerical instabilities arise when close to event horizon curvature singularities.

Robust adaptive schemes are being explored to address these issues.

IV. CONCLUSION

In this work, we proposed a novel integration of classical and advanced Data Structures and Algorithms (DSA) into the simulation pipeline of black hole environments, aiming to enhance computational efficiency, scalability, and physical fidelity. The fusion of octrees, KD-Trees, and hash-based memory-accelerated structures with general relativistic simulation algorithms has demonstrated substantial performance gains across multiple dimensions, including runtime reduction, memory optimization, and cache utilization.

Our experiments, underpinned by rigorous performance metrics and comparative analysis, reveal that algorithmically-informed simulations—particularly those powered by spatial partitioning trees—can outperform traditional physics-driven solvers in both speed and adaptability. These hybrid architectures offer new computational avenues for solving Einstein's field equations, simulating event horizons, and modeling relativistic particle interactions around singularities, especially when scaled to high-resolution grids or large-scale astrophysical scenarios.

Moreover, by embedding DSAs within numerical relativity engines, we introduced a paradigm shift in how space-time curvature and gravitational interactions can be encoded and navigated. This rethinking opens new research frontiers, including dynamic memory optimization in tensor field solvers, and metaheuristic approaches to evolve simulation configurations based on feedback from data structures themselves.

The implications of our framework are twofold: (1) it sets the stage for ultra-fast simulations that remain physically grounded, and (2) it offers an adaptable foundation for integrating AI, quantum-enhanced computing, and distributed systems into future black hole modeling frameworks. Our results strongly advocate for a cross-disciplinary approach where computational theory, astrophysics, and software engineering coalesce.

Future work will aim to incorporate neurosymbolic reasoning to detect gravitational wave signatures, leverage distributed hash tables across GPU clusters, and employ reinforcement learning agents to evolve optimal simulation paths. With the accelerating demand for high-fidelity cosmic simulations—driven by missions like LISA and the Event Horizon Telescope—our DSA-augmented approach offers a timely, scalable, and intelligent leap forward in computational astrophysics.

References

[1] B. P. Abbott et al., "Observation of Gravitational Waves from a Binary Black Hole Merger," *Phys. Rev. Lett.*, vol. 116, no. 6, p. 061102, 2016.

[2] Event Horizon Telescope Collaboration, "First M87 Event Horizon Telescope Results. I. The Shadow of the Supermassive Black Hole," *Astrophys. J. Lett.*, vol. 875, no. 1, p. L1, 2019.

[3] M. Alcubierre, *Introduction to 3+1 Numerical Relativity*, Oxford University Press, 2008.

[4] A. V. Goldberg and R. E. Tarjan, "A New Approach to the Maximum Flow Problem," *J. ACM*, vol. 35, no. 4, pp. 921–940, 1988.

[5] Pomarède, D., Fidaali, Y., Teyssier, R. (2008, August). Visualization of Hybrid, N-body and Octree-based Adaptive Mesh Resolution Parallelized Simulations. In 2008 Fifth International Conference on Computer Graphics, Imaging and Visualisation (pp. 295-304). IEEE.

[6] H. Samet, *Foundations of Multidimensional and Metric Data Structures*, Morgan Kaufmann, 2006.

[7] F. Pretorius, "Evolution of Binary Black Hole Spacetimes," *Phys. Rev. Lett.*, vol. 95, p. 121101, 2005.

[8] El Skaf, R., Palacios, P. (2022). What can we learn (and not learn) from thought experiments in black hole thermodynamics?. Synthese, 200(6), 434.

[9] Shapiro, S. L., Teukolsky, S. A. (2024). Black holes, white dwarfs and neutron stars: the physics of compact objects. John Wiley Sons.

[10] Alcubierre, M. (2008). Introduction to 3+ 1 numerical relativity (Vol. 140). OUP Oxford.

[11] Pretorius, F. (2005). Evolution of binary black-hole spacetimes. Physical review letters, 95(12), 121101.

[12] Pretorius, F., Choptuik, M. W. (2006). Adaptive mesh refinement for coupled elliptic-hyperbolic systems. Journal of Computational Physics, 218(1), 246-274.

[13] Venkataraman, S., Benger, W., Long, A., Jeong, B., Renambot, L. (2006, November). Visualizing hurricane katrina: Large data management, rendering and display challenges. In Proceedings of the 4th international conference on Computer graphics and interactive techniques in Australasia and Southeast Asia (pp. 209-212).

[14] Baker, J. G., Centrella, J., Choi, D. I., Koppitz, M., van Meter, J. (2006). Binary black hole merger dynamics and waveforms. Physical Review D—Particles, Fields, Gravitation, and Cosmology, 73(10), 104002.

[15] Clough, K., Figueras, P., Finkel, H., Kunesch, M., Lim, E. A., Tunyasuvunakool, S. (2015). GRChombo: Numerical relativity with adaptive mesh refinement. Classical and Quantum Gravity, 32(24), 245011.

[16] Maule, M., Maciel, A., Nedel, L. (2010, August). Efficient collision detection and physics-based deformation for haptic simulation with local spherical hash. In 2010 23rd SIBGRAPI Conference on Graphics, Patterns and Images (pp. 9-16). IEEE.

[17] Komissarov, S. S. (2005). Observations of the Blandford–Znajek process and the magnetohydrodynamic Penrose process in computer simulations of black hole magnetospheres. Monthly Notices of the Royal Astronomical Society, 359(3), 801-808.

[18] Baiotti, L., Rezzolla, L. (2017). Binary neutron star mergers: a review of Einstein's richest laboratory. Reports on Progress in Physics, 80(9), 096901.

[19] Cook, G. B., Pfeiffer, H. P. (2004). Excision boundary conditions for black-hole initial data. Physical Review D—Particles, Fields, Gravitation, and Cosmology, 70(10), 104016.

[20] Hawley, J. F., Smarr, L. L., Wilson, J. R. (1984). A numerical study of nonspherical black hole accretion. I Equations and test problems. Astrophysical Journal, Part 1 (ISSN 0004-637X), vol. 277, Feb. 1, 1984, p. 296-311. Research supported by the US Department of Energy., 277, 296-311.

[21] Barber, J., Antonini, F. (2025). Formation and evolution of binary black holes in N-body simulations of star clusters with up to two million stars. Monthly Notices of the Royal Astronomical Society, 538(2), 639-658.